

A Deep Neural Network Based Predictive Control Strategy for High Frequency Multilevel Converters

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Abstract—This paper investigates a new deep neural network (DNN) based predictive control strategy for high frequency multilevel converters. Compared to the conventional model predictive control (MPC) strategy, the new strategy significantly reduces computational burden, and improves calculation accuracy of mathematical models for power converters. A 5-level flying capacitor converter is used as a case study to compare these two strategies. It is found that the DNN based control strategy offers effective control of the load current and capacitor voltages with a much reduced computational requirement. The new concept is validated by both numerical analysis and hardware-in-loop experimental results.

Keywords — MPC; FCMC; neural network

I. INTRODUCTION

Nowadays, power converters play an important role in renewable energy system, energy storage system, drive system of electric vehicle .etc. There have been many literatures about how to control various power converters efficiently and steadily [1-2]. A common control strategy is linear proportional-integral (PI) which offers good performance. However, it has been shown that the nonlinear control strategies can improve the transient and steady state characteristics of power electronic systems compared with conventional linear approaches. One promising choice of nonlinear control methods for power converters is model predictive control (MPC). In recent years, MPC is gaining considerable attention for power converter and drive applications because it can easily handle multiple control objectives, system constraints, and nonlinearities in a very intuitive way [3]. The MPC control strategy uses a power converter model to provide the control action sequence for the system by optimizing a user-defined cost function. A major challenge of MPC is the exponential increase of computational burden when a small sampling time is applied to complex converter topologies with a large number of switching states, even with the rapid advancement of microprocessor technology. Another challenge of MPC is the calculation accuracy of the converter models especially with relatively large sampling times since the differential equations are usually approximated with the finite difference method [3].

Multilevel converters, especially those using high speed wide bandgap (WBG) semiconductor devices, are attractive choices for further power electronics applications [4]. Flying capacitor multilevel converters (FCMC) using GaN switches demonstrated a record-setting power density and efficiency in [4]. With increasing voltage level, the number of FCMC controlled variables increases significantly, raising serious challenges to the control strategy development. While MPC seems to be the right choice [5][6], the aforementioned computational load and accuracy remain two major technical barriers.

In this paper, we develop a new deep neural network (DNN) based predictive control strategy for high frequency multilevel converters. This new strategy offers two attractive characteristics: lower computational cost and higher calculation accuracy of the mathematical models for the converter. Section II shows a typical FCMC converter and its mathematical models for MPC. Section III introduces the new DNN based predictive control strategy, neural network design, neural network testing, and computational cost analysis. A 5-level FCMC is used as a case study to compare these two control strategies. Experimental validation using control hardware in the loop (CHIL) simulation is discussed in Section IV.

II. MULTILEVEL CONVERTER

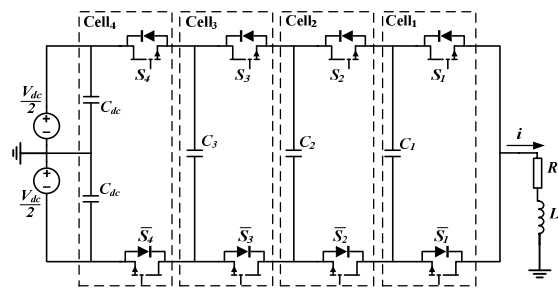


Fig. 1. Single phase five-level FCMC converter used as a case study of the proposed strategy.

We use a single-phase five-level FCMC converter as a case study in this work, as shown in Fig. 1. Each modular cell is composed of one capacitor C_i and two switches S_i and

$\bar{S}_i$, which work in a complementary manner. To develop mathematical models for the FCMC, we have

$$V_o^{k+1} = S_n^k(V_{dc} - V_{c_{n-1}}^k) + \dots + S_{i+1}^k(V_{c_{i+1}}^k - V_{c_i}^k) + \dots + S_1^k V_{c_1}^k \quad (1)$$

$$i^{k+1} = i_k e^{-TR/L} + (1 - e^{-TR/L}) V_o^{k+1} / R \quad (2)$$

$$V_{c_i}^{k+1} = V_{c_i}^k + \frac{T}{2C_i} (i^k + i^{k+1})(S_{i+1}^k - S_i^k) \quad (3)$$

where $V_{c_i}^k$ is the voltage drop on C_i , i^k is the load current, V_o^{k+1} is the predictive output voltage at the sampling instant $k+1$, i^{k+1} is the predictive load current at the sampling instant $k+1$. $S_i, \bar{S}_i \in \{0, 1\}$. If $S_i=0$, the top switch of i cell is off, and vice versa.

In this paper, the cost function is given by tracking error of the load current and the voltage drop on each flying capacitor in the converter.

$$J = (i^* - i^{k+1})^2 + \sum_{i=1}^N \lambda_{c_i} (V_{c_i}^* - V_{c_i}^{k+1})^2 \quad (4)$$

Where i^* is the reference value of the load current, its value is determined by the output power. $V_{c_i}^*$ is the reference value of the voltage drop on flying capacitor (C_i) which is in descending order from DC bus to load. $V_{c_i}^{k+1}$ is the predictive value of the voltage drop on flying capacitor at the sampling instant $k+1$. Weight coefficient λ_{c_i} is used to weigh the proportion of each status variable. One can obtain the optimal control sequence which is corresponding to the minimum cost function to control the various switches.

At k step, we assume that the status variables of FCMC system is $x^k = [V_o^k, V_{cN}^k, V_{c(N-1)}^k, \dots, V_{c1}^k]^T$ and the input is $u^k = [S_{cN}^k, S_{c(N-1)}^k, \dots, S_{c1}^k]^T$. Then the status variables of system at $k+1$ step (x^{k+1}) can be calculated by the mathematical models of FCMC:

$$x^{k+1} = f(x^k, u^k) \quad (5)$$

The cost function J contain the reference value and x^{k+1} . Combined with equ. 5, the J in $k+1$ step can be expressed as:

$$J^{k+1} = g(x^*, x^{k+1}, \lambda_{c_i}) = h(x^k, u^k) \quad (6)$$

Afterwards the optimal input signals can be calculated:

$$u^k = \arg \min h(x^k, u^k) \quad (7)$$

III. DNN BASED PREDICTIVE CONTROL STRATEGY

The essence of a predictive control strategy to map its input (The status variables sampled in k step) to its output (The optimal switch signals in k step). The certain output and system states is corresponding to a group of optimal control signals. It is known that a deep neural network can describe complex nonlinear relations with sufficient precision. Hence, a deep neural network is first built and trained offline by the input and output data based on MPC

operation of the converter but without the usual computational time constraint.

The training samples are made up by two part. The input vector X consist of the status variables of FCMC system:

$$X = [I_L, (I_L^* - I_L), (V_{c3}^* - V_{c3}), (V_{c2}^* - V_{c3}), (V_{c1}^* - V_{c3})]^T \quad (8)$$

To make training process convenient, the samples are normalized and mapped to a unit cube I_5 ($I_5 \in (0, 1)^5$) in five dimensional state space.

The optimal switch function $Y = [S_1, S_2, S_3, S_4]^T$ calculated by equ. 7 is selected as the output vector. A DNN with small number of training samples cannot well describe the complex nonlinearity of model predictive control. However, too much samples may cause overfitting in training process, resulting in low accuracy too. In this paper, we use 100 thousands samples for training. The generated weight matrixes are then used for calculating the control signals for the actual operation of the FCMC in real time, which requires much less computational resources than the conventional MPC model calculation. Note that the MPC differential equations used in the offline training are solved much more accurately with a finer discretization method than the conventional MPC since computational speed is not a constraint in offline training. In some papers, in order to reduce computational burden, look-up table in predictive control is used [7]. However, in some complex topologies like multilevel converter. The look-up table method need to record various system status, resulting in high cost of storage resource. The neural network, on the other hand, take into consideration both the computational burden and storage resource. With DNN, the controller only need to save weight matrix and deal with multiplication, addition and logical operation which can be easily handled by FPGA.

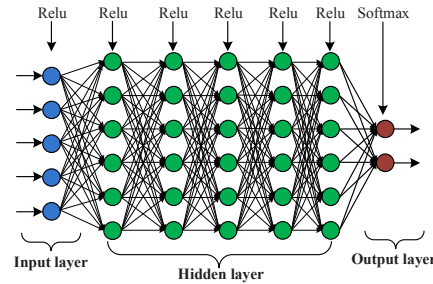


Fig. 2. Deep neural network designed for predictive control of FCMC.

Compared with artificial neural network (ANN), DNN has a flatter structure, resulting in higher precision. Fig. 2 shows the architecture of the proposed deep neural network. The DNN adopts a typical back-propagation neural network structure with 7 layers: one input layer with five neurons, one output layer with two neurons, and five hidden layers with 6 neurons in each layer. In literatures, ANN based power electronic controllers often use nonlinear function (e.g. sigmoid functions) as its activation function [8], which need to be piecewise linearized in hardware implementation and the computational accuracy will significantly degrade when computational burden is high. In this work, we adopt Relu function as the activation function. As shown in Fig. 3, different from the commonly used sigmoid function, Relu function is piecewise linear and easy to implement. A softmax function is adopted in the output layer. Because of the monotonically increasing characteristic of the softmax

function, the microprocessor just needs to calculate the net output values of the output layer and find out the maximum value in hardware implementation. Therefore, the computation of the predictive control becomes highly parallel linear calculation and logical operation which can be easily handled by a FPGA. In the conventional MPC, restricted by the high computational cost, the state equations must be discretized as the system models and the prediction step is limited to a single step. These restrictions degrade the control performance. In DNN based predictive control, the state equations of the converter system are directly chosen as the control models, and the prediction step can be multistep. The DNN is trained offline with the input and output data of the controlled system, but the online control only needs to process the saved weight matrixes. In this way, computation speed and model precision of the controller with DNN based predictive control strategy are both improved.

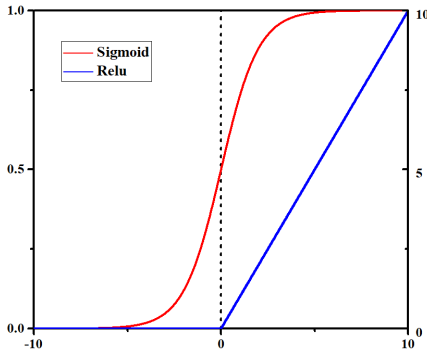


Fig. 3. Relu function and Sigmoid function.

TABLE I. Comparison of computational requirement of FCMC as number of calculation within one sampling period

	# of calculation in one sampling cycle	
Level	n=4	n=N
MPC	15360	$3(2^{N-1})^3(3N-2)$
DNN based PC	604	$3\epsilon_N M^2(L-1)$

There are two approaches to implement the DNN control strategy. The first approach is to assign a particular neural network to each switch cell while the second approach is to generate a bigger deep neural network for the entire group of all the switch cells. The accuracy of the second approach is higher with a penalty of higher training cost. However, for the single big DNN, we can generate several networks trained by different data and compare their output for the cost functions. Subsequently, the output corresponding to the minimum cost function is chosen as our control signal vector. If the accuracy of one DNN is $\alpha_1, \alpha_2, \alpha_3 \dots \alpha_N$ which may be not very high, the accuracy of the group of DNN can be improved to $1 - \prod_{n=1}^N (1 - \alpha_n)$. Note that training

a neural network with high accuracy is much more complex than training one with lower accuracy. In this way, we can obtain neural network group with considerably high accuracy and much reduced training cost.

The computational cost of the MPC strategy and DNN based predictive control are compared in Table I. ϵ_N is a coefficient increases slowly with the rising of N. When N is 7, the typical value of ϵ_N is 1.8. M is the number of neuron in each hidden layer while L is the quantity of hidden layer. From Table I, it is seen that the computational burden in terms of number of calculation performed within one sampling period is significantly reduced with the proposed DNN based predictive control.

As described in the previous section, we use the high-precision differential equations as the system model of a single phase flying capacitor five-level converter and calculate the corresponding optimal switching states of each switch cell for various system states. Then the above data are taken as our training samples. We build four neural networks: A, B, C, D and use them to control S4, S3, S2, S1, respectively. The accuracy of the four neural networks is estimated in Table II, all above or near 90%.

TABLE II. Accuracy estimate of DNN

DNN	Controlled Switch	Accuracy
A	S4	86%
B	S3	87%
C	S2	89%
D	S1	91%

IV. EXPERIMENTAL RESULTS



Fig. 4. Control hardware in the loop (CHIL) setup

TABLE III. Circuit parameters of the 5L FCMC prototype

Parameters	Value
V_{dc}	400V
Switching device	GaN FET
Flying capacitor	100 μ F
Sampling frequency	100kHz
Load	10 Ω /1mH

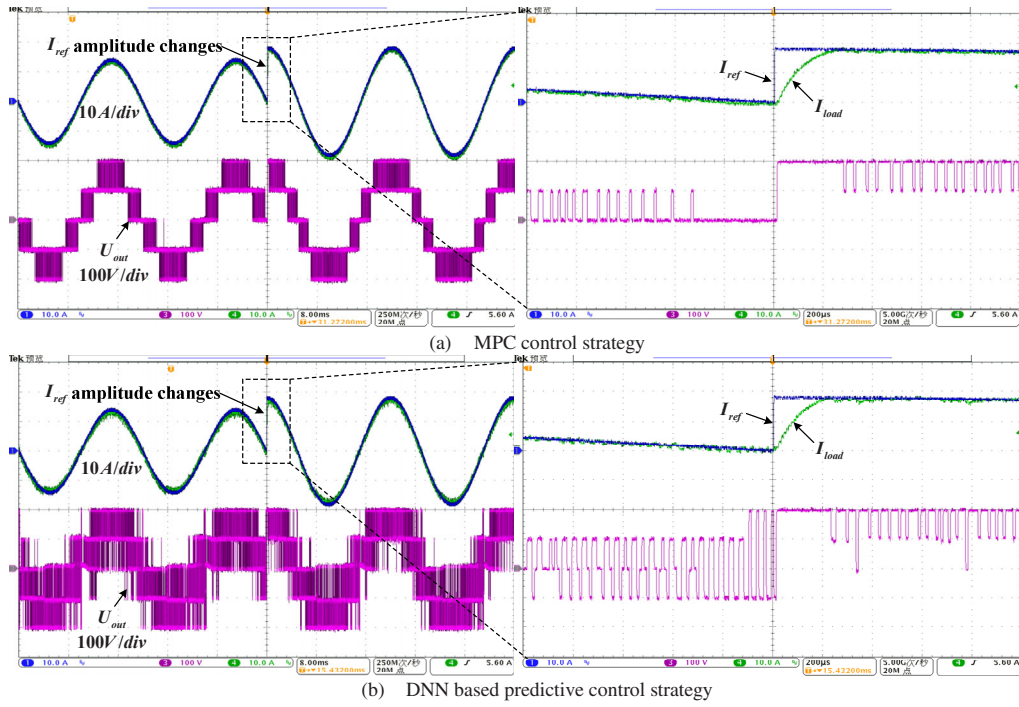


Fig. 5. Load current and output voltage waveforms of FCMC with MPC and DNN control strategies.

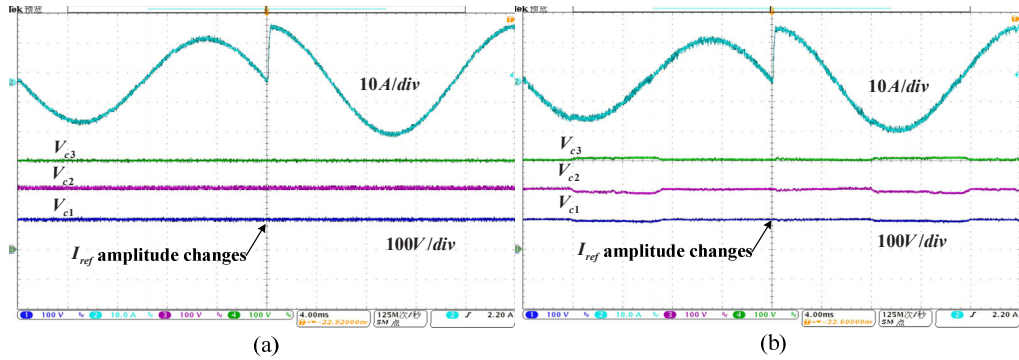


Fig. 6. Capacitor voltage waveforms of the FCMC with (a) MPC and (b) DNN control strategies.

We use control hardware in the loop (CHIL) simulation to validate our proposed DNN based predictive control strategy experimentally. The setup of our CHIL is shown in Fig. 4. We choose the single-phase 5-level FCMC prototype in our work. We use a FPGA (Artix-7 100T AX7102) as the hardware controller. The system parameters are summarized in Tables III.

Fig. 5 shows the load current I_{load} as well as the output voltage of the FCMC which is controlled with the MPC and DNN based predictive control strategies, respectively. To test the robustness of the predictive control strategies, we purposely change the amplitude of reference current I_{ref} from 14A to 18A during operation. It is observed that the new DNN based strategy controls the load current transient as effectively as the conventional MPC strategy even with a much less computational burden. Fig. 7 compares the THD of the load current of the FCMC in steady state operation at 18A with the MPC and DNN based predictive control strategy, respectively. The load current THD is 2.92% and 3.53% for the MPC and the DNN based predictive control strategy, respectively, quite low in both cases.

Fig. 6 shows the capacitor voltage waveforms of the FCMC with MPC and DNN based predictive control strategy, respectively. It can be observed that the capacitor voltages remain stable during load current transient in both cases, but with the DNN method requiring much less computational resources. The experiment results validate the performance of our new predictive control strategy.

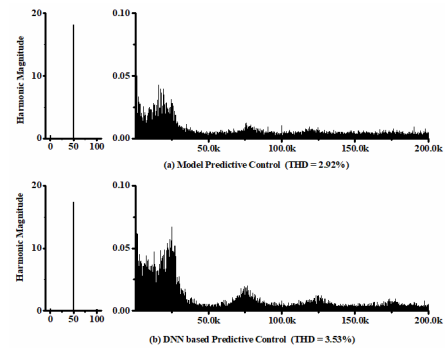


Fig. 7. Comparison of load current THD.

V. CONCLUSION AND FUTURE WORK

This paper reports a new DNN based predictive control strategy for high frequency multilevel converters. Compared to the conventional model predictive control (MPC) strategy, the new strategy significantly reduces computational burden, and improves calculation accuracy of the mathematical models for power converters. Due to the greatly reduced computational burden, the control strategy becomes more flexible since it can be generalized from single step to multistep. The modeling and experimental results on the 5-level FCMC case study reveal that the DNN based predictive control strategy offers effective control of the load current and capacitor voltages with a much reduced computational requirement and improved accuracy in comparison to the conventional MPC control. The new concept can be used in broader applications requiring predictive control, offers significant improvements in processing speed and model precision.

Although the differential equations of DNN based predictive control used in the offline training are solved much more accurately with a finer discretization method than the conventional MPC. However, it can be seen from Fig. 5, 6&7 that the performances of five-level flying capacitor converter with DNN based predictive control and conventional MPC have little difference with each other. This is because in this paper, we build four neural networks to control the four switch cells in five-level FCMC, respectively. In this way, the training cost of neural network is effectively reduced. But, on the other hand, the accuracy of the whole controller is reduced to the product of the accuracy of the four DNNs. This will affect the control performance to some extent. In further work, we plan to generate a bigger deep neural network as described in above section for the entire group of all the switch cells. With this

method, the nonlinearity of MPC can be better described with much lower computational burden.

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